

A model 97
Microlight
flying above
Mojave Airport.

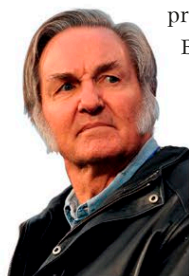


Great Manufacturers Scaled Composites

As the manufacturer of both the first aircraft to fly nonstop, nonrefueled around the world and the world's first private manned space vehicle, Scaled Composites is an unusual company, built by a remarkable man. The story of Burt Rutan and Scaled Composites is one of innovation and genius.

SCALED COMPOSITES was the creation of Californian aircraft and spacecraft designer, Burt Rutan. It is no exaggeration to say that Rutan transformed light aviation. In fact, five of his designs are on display at the Smithsonian's National Air and Space Museum, Washington, DC. Starting in the early 1970s, he designed a series of aircraft, each more unconventional than its predecessor, while setting new standards in performance. From selling plans for small homebuilt aircraft to designing a spacecraft, Rutan and Scaled Composites have truly "pushed the envelope."

Born in 1943 and christened Elbert (although known as Burt), Rutan became interested in aviation at a young age and became an aeronautical engineer. Following a stint as a project engineer at the Edwards Air Force Test Center in Mojave, his first professional involvement in the homebuilt aircraft world was assisting Jim Bede in the design of the Bede BD-5J single-seater jet.



Burt Rutan
(1943-)

In 1974, he returned to California and formed the Rutan Aircraft Factory (RAF) at Mojave Airport. RAF's first project was to market plans of Burt's Model 27 VariViggen,

which was named after the Saab Viggen—a jet fighter it "vari" much resembled (bad puns would often figure in early Rutan aircraft names). An improved VariViggen followed, as Rutan looked at refining the wing to

increase performance. This model introduced a new construction technique—the outer wing panels being made from urethane foam skinned with fiberglass. Rutan's next project—the VariEze (so named because it was very easy to build) was a huge success. Burt's brother Dick even set a closed-course distance record for sub-1,100 lb (-500 kg) aircraft of 1,637 miles (2,656 km) in the prototype.

Over the next decade, Rutan designed several more innovative types. These included the twin-engine Model 40 Defiant, the Solitaire



motorglider, the STOL Grizzly, asymmetric Boomerang, and Proteus high-altitude research vehicle. Rutan's skills as a designer were soon noticed outside the homebuilder community, and in 1979 NASA's Dryden Flight Research Center commissioned him to complete the design work on a small, jet-powered pivoting-wing research aircraft, the AD-1.

In 1983 the Rutan Aircraft Factory was reorganized as Scaled Composites, because Rutan wanted to get out of the kitplane business and concentrate on research and development work for advanced concept aircraft. An early project was an 85 percent scale proof-of-concept (POC) aircraft for the Beech Starship—a business class, canard-configured, twin "pusher." Beech also bought Scaled Composites around this time, although Rutan was still very much in charge. Scaled also

Aviation memorabilia

This postcard showcases some of Scaled Composites' spaceplanes, and is signed by Burt Rutan and his brother, Dick.

designed Voyager, an airplane that could fly around the world nonstop and nonrefueled. Flown by Dick Rutan and Jeana Yeager, Voyager's record-setting around-the-world flight started from Edwards Air Force Base, California, on December 14, 1987, and arrived back there on December 23.

In March 2004, Scaled built a new aircraft for Sir Richard Branson and adventurer Steve Fossett. GlobalFlyer (Model 311) eventually made two solo nonrefueled around-the-world flights piloted by Fossett, smashing Voyager's record of 24,987 miles (40,212 km).

Scaled has not confined its activities to fixed-wing aircraft. In 1993, the company took on the development of a tilt-rotor, unmanned, aerial vehicle



VariViggen
Rutan began working on the design of the VariViggen while a student at Cal-Poly in the early 1960s. Three years after graduation, he started building the prototype in his garage.